

Event-Based Colour Response Analysis and Pseudo-Colour Reconstruction

Technical Report

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Abstract

This project investigates whether colour-related information can be extracted from a monochrome event camera using controlled sequential red, green and blue illumination.

Unlike conventional frame-based cameras, an event camera does not directly record absolute pixel intensity. Instead, individual pixels asynchronously generate events in response to changes in logarithmic brightness. Conventional RGB information is therefore not directly available from the raw event stream.

A Python processing pipeline was developed to analyse Prophesee-style DAT recordings. The system performs memory-efficient event decoding, automatic localisation of an initial illumination stage, fixed-period temporal alignment of subsequent RGB stages, interactive region-of-interest selection, ON/OFF event-response measurement, repeatability analysis, relative RGB response estimation and pseudo-colour event visualisation.

A central design choice is that only the initial illumination phase is detected from the event stream. Subsequent RGB measurement windows are generated from a configured timing model rather than independently shifted toward nearby event peaks. This reduces the risk of interpreting PWM-induced event bursts as separate physical illumination onsets.

For each region of interest, the colour feature is obtained by selecting the maximum ON-event density observed across repeated red, green and blue measurements. These three responses are sum-normalised to form a relative RGB response vector. The vector is treated as a sensor–illumination–surface response signature rather than as calibrated surface RGB.

The resulting framework provides a reproducible basis for studying colour-related event responses and for future work involving hardware synchronisation, spectral calibration, RGB ground truth, motion compensation and multimodal robotic perception.

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1 Introduction

Event cameras are bio-inspired visual sensors that operate differently from conventional frame-based cameras.

A conventional camera records complete images at fixed time intervals. An event camera instead allows each pixel to operate asynchronously. When the local change in logarithmic brightness exceeds an internal contrast threshold, the sensor generates an event.

A single event can be represented as

$$e_i = (x_i, y_i, t_i, p_i), \quad (1)$$

where

- x_i and y_i represent pixel coordinates,
- t_i represents the timestamp,
- p_i represents the event polarity.

This sensing principle provides high temporal resolution, low latency, high dynamic range and reduced motion blur.

However, a monochrome event sensor does not directly encode conventional RGB colour.

The central idea explored in this project is to illuminate the same surface sequentially with red, green and blue light and compare the event response generated under each illumination channel.

Conceptually,

1. illuminate the scene using red light,
2. measure the event response,
3. repeat under green light,
4. repeat under blue light,
5. compare the three responses.

If different surfaces produce different relative responses under these illumination channels, the measured event activity may contain useful colour-related information.

The aim is therefore not to reconstruct conventional calibrated RGB intensity directly. Instead, the project develops an experimental framework for measuring and visualising relative colour-related responses from a monochrome event camera.

2 Research Questions

The project considers the following questions:

1. How does a monochrome event camera respond to controlled red, green and blue illumination?
2. Can event responses under different illumination channels provide a meaningful relative RGB response vector?
3. How can an asynchronous event stream be aligned with the illumination sequence?
4. Why can the largest event-count peaks be unreliable indicators of physical illumination onset?
5. How should event responses from differently sized spatial regions be compared?
6. How repeatable are the measured responses across multiple RGB cycles?
7. What prevents the resulting response vector from being interpreted as calibrated RGB colour?
8. How could this approach be extended toward more general event-based and multimodal perception?

3 Experimental Assumptions

The software relies on several assumptions about the data-acquisition experiment.

The illumination sequence is assumed to follow the known order

$$R \rightarrow G \rightarrow B.$$

For three repetitions,

$$R_1, G_1, B_1, R_2, G_2, B_2, R_3, G_3, B_3.$$

The first automatically detected illumination stage is interpreted as R_1 .

This colour identity is not determined directly from the event stream. The data can provide evidence that an illumination-related transition occurred, but without an external colour reference they cannot determine whether an unknown first light was red, green or blue.

Therefore,

The interpretation of the first detected illumination stage as R_1 depends on the known acquisition sequence.

The temporal configuration must also correspond to the physical experiment. Values specified in software describe the assumed acquisition timing but do not themselves guarantee that the LEDs produced the exact corresponding optical waveform.

When direct timing measurements are available, they should therefore be preferred over nominal software values.

4 DAT Event Decoding

4.1 Input Representation

The current implementation processes Prophesee-style DAT recordings.

The software expects CD events stored using 8-byte event records. Each record consists of two 32-bit words:

- a timestamp word,
- an encoded data word containing coordinates and polarity.

The decoder reconstructs

$$(x, y, t, p)$$

for each event.

Before processing, the software verifies information including:

- DAT version where available,
- event type,
- event size,
- whether the remaining file length contains complete events,
- chronological timestamp ordering,
- sensor width and height.

These checks reduce the risk of silently analysing data using an incorrect file-format assumption.

4.2 Memory-Efficient Processing

Large event recordings may contain millions of events.

The implementation therefore avoids loading the complete recording into memory simultaneously.

The default chunk size is

```
1 chunk_events = 1_000_000
```

The basic processing pattern is:

```
1 for timestamps, x, y, polarity in iter_event_chunks(...):
2     # Process one section of the event stream
3     ...
```

Only the currently required event block and relatively small temporal histograms need to remain in memory.

Timestamp-based binary search is also used when the program needs to seek to a specific temporal position in the DAT file.

5 Temporal Synchronisation Problem

Temporal alignment is critical because the RGB response depends directly on which events are assigned to each illumination stage.

A simple approach would be to locate the largest event-count peaks and associate each one with a red, green or blue illumination window.

During development, this was found to be unreliable.

PWM-driven illumination and other rapid changes in the light output can produce several strong event bursts within a single physical illumination stage.

Consequently,

$$\text{maximum event count} \neq \text{physical illumination onset.} \quad (2)$$

A strong event peak may instead represent:

- a PWM transition,
- a later stage of an illumination ramp,
- a brightness decrease,
- local object response,
- motion-related activity.

Independently moving every RGB stage toward the strongest nearby event peak could therefore produce visually plausible alignment while no longer representing the actual experimental timing.

The implemented strategy instead separates the problem into two steps:

1. automatically identify one plausible initial illumination phase;
2. calculate all subsequent RGB windows using one fixed timing model.

6 Automatic First-Flash Detection

6.1 Whole-Recording Scan

Before fine timing localisation, the retained recording is scanned from beginning to end.

This is important because the recording may contain a long low-activity period before the RGB experiment begins.

A silent interval is not treated as the end of the search.
The whole-recording scan provides:

- total recording duration,
- total ON and OFF activity,
- coarse temporal activity,
- large inter-event gaps,
- timestamp-order validation.

The coarse histogram is used for diagnostic purposes only.
It is not used as the final flash detector.

6.2 Fine Temporal Search

Fine timing detection uses a smaller temporal bin.

With the current default configuration,

```
1 flash_detection_bin_us = max(
2     1,
3     min(10, FADE_DURATION_US // 20)
4 )
```

For a 1000 μ s fade duration, this produces a 10 μ s detection bin.

Candidate first-onset times are evaluated according to whether the complete expected RGB schedule produces event activity consistent with the configured light and dark intervals.

6.3 ON-Event Contrast Score

For one candidate illumination stage, let

$$N_{\text{light}}$$

represent ON-event activity inside the expected illumination interval and

$$N_{\text{dark}}$$

represent ON-event activity inside a later predicted dark interval.

A simplified form of the standardised ON contrast score is

$$S_{\text{ON}} = \frac{N_{\text{light}} - \alpha N_{\text{dark}}}{\sqrt{N_{\text{light}} + \alpha^2 N_{\text{dark}}}}, \quad (3)$$

where α compensates for differences between the lengths of the two measurement intervals.

The implementation also evaluates OFF-event activity near the expected end of the illumination stage.

These values provide statistical evidence for or against a candidate timing configuration.

They are not calibrated probabilities.

6.4 Sequence-Level Validation

A first-onset candidate is accepted only if sufficient portions of the complete fixed RGB sequence contain supporting event evidence.

The current configuration includes:


```

1 timing_min_flash_score = 2.5
2 timing_min_sequence_score = 5.0
3 timing_min_supported_fraction = 2 / 3
4 timing_max_startup_score = 3.0

```

The algorithm also evaluates activity before the candidate onset so that an already-active section of the recording is less likely to be mistaken for the experiment start.

Passing these statistical checks means that the event activity is consistent with the configured RGB sequence.

It does **not** mean that exact hardware synchronisation has been established.

7 Fixed RGB Temporal Alignment

Once a first onset t_0 is detected, subsequent RGB stages are generated from the configured period.

For flash index j ,

$$t_j = t_0 + jT + \left\lfloor \frac{j}{3} \right\rfloor T_{\text{extra}}, \quad (4)$$

where

- T is the adjacent illumination-stage period,
- T_{extra} represents an optional additional delay after a complete RGB cycle.

The nominal period is calculated as

$$T = T_{\text{fade}} + T_{\text{peak}} + T_{\text{dark}}, \quad (5)$$

unless an experimentally measured period is explicitly provided.

The current default values are

$$T_{\text{fade}} = 1000 \mu\text{s}, \quad (6)$$

$$T_{\text{peak}} = 0 \mu\text{s}, \quad (7)$$

$$T_{\text{dark}} = 500 \mu\text{s}. \quad (8)$$

Therefore,

$$T = 1500 \mu\text{s}. \quad (9)$$

If the physical adjacent-stage period has been measured, it can instead be specified through

```

1 MEASURED_FLASH_PERIOD_US = ...

```

For example,

```

1 MEASURED_FLASH_PERIOD_US = 1268

```

would use

$$T = 1268 \mu\text{s}.$$

The important methodological constraint is that later stages are not independently shifted toward their local event maxima.

7.1 Timing Drift and Interpretation

A correct initial onset does not guarantee that every later response window remains physically aligned.

If the configured period differs from the true hardware period by ΔT , the timing error after j stages is approximately

$$\Delta t_j \approx j\Delta T. \quad (10)$$

A small period error can therefore accumulate across repeated RGB cycles.

If the first cycle appears aligned but later cycles progressively shift relative to the observed event response, the period should be investigated before changing the individual flash positions.

The automatic timing test should therefore be interpreted as a **consistency check**, not as proof of exact camera–illumination clock synchronisation.

8 Response Measurement Windows

The quantitative event-response window can begin at or shortly after the predicted illumination onset.

For illumination stage j ,

$$W_j = [t_j + d, t_j + d + w), \quad (11)$$

where

- d is the response delay,
- w is the response-window duration.

The current default settings are:

```
1 response_delay_us = 0
2 response_window_us = FADE_DURATION_US
```

Thus, with the default configuration, the complete nominal fade interval is used for response measurement.

For example,

```
1 response_delay_us = 100
2 response_window_us = 500
```

would measure a 500 μs interval beginning 100 μs after each predicted onset.

The configuration must satisfy

$$d + w \leq T, \quad (12)$$

so that one response interval does not extend into the next illumination stage.

9 Region-of-Interest Analysis

9.1 Reference Event Image

The raw event stream is asynchronous and therefore does not naturally provide a conventional image for manual ROI selection.

A temporary accumulated event frame is generated so that the user can identify the target surfaces.

The reference display is used only for ROI selection.

Changing its accumulation duration, brightness or visual event size does not modify:

- the detected initial illumination phase,
- the fixed RGB timing schedule,
- the response measurement windows,
- the quantitative event counts.

This separation prevents display choices made during ROI selection from changing the underlying measurement.

9.2 Polygonal ROIs

The program supports an arbitrary positive number of polygonal regions of interest.

When

```
1 num_rois = None
```

the number is requested at run time.

Each polygon is then defined interactively using the reference event image.

The first ROI is normally treated as the reference surface when the optional material-contrast display is used.

9.3 Nested ROIs

A defect or test region may be completely contained inside a larger reference region.

If both masks were analysed independently, events inside the inner ROI would also be counted as part of the outer surface.

The default setting is therefore

```
1 nested_roi_mode = "EXCLUDE_CONTAINED"
```

If ROI B is completely contained inside ROI A , the effective statistical mask is

$$M_A^{\text{effective}} = M_A \setminus M_B. \quad (13)$$

The inner ROI remains unchanged.

This allows a contained defect and the surrounding reference surface to be measured separately.

If independent overlapping regions are required, the mode can be changed to

```
1 nested_roi_mode = "INDEPENDENT"
```

Partial overlaps that do not correspond to strict containment are not automatically removed.

10 Event-Response Measurement

For ROI r , illumination colour c and repetition k , the program measures

$$N_{r,c,k}^{\text{ON}}, \quad (14)$$

$$N_{r,c,k}^{\text{OFF}}. \quad (15)$$

The total number of events is

$$N_{r,c,k}^{\text{TOTAL}} = N_{r,c,k}^{\text{ON}} + N_{r,c,k}^{\text{OFF}}. \quad (16)$$

Because raw event counts depend on ROI size, the measurements are normalised by the effective ROI area A_r .

For ON events,

$$D_{r,c,k}^{\text{ON}} = \frac{N_{r,c,k}^{\text{ON}}}{A_r}. \quad (17)$$

Similarly,

$$D_{r,c,k}^{\text{OFF}} = \frac{N_{r,c,k}^{\text{OFF}}}{A_r}, \quad (18)$$

and

$$D_{r,c,k}^{\text{TOTAL}} = \frac{N_{r,c,k}^{\text{TOTAL}}}{A_r}. \quad (19)$$

Area-normalised density provides a more meaningful comparison between regions containing different numbers of pixels.

11 Repeatability Analysis

Measurements from repeated RGB cycles are retained individually.

For repeated event densities

$$D_1, D_2, \dots, D_n,$$

the mean is

$$\bar{D} = \frac{1}{n} \sum_{k=1}^n D_k. \quad (20)$$

The sample standard deviation is

$$s_D = \sqrt{\frac{\sum_{k=1}^n (D_k - \bar{D})^2}{n - 1}}. \quad (21)$$

The coefficient of variation is calculated as

$$\text{CV} = 100 \frac{s_D}{\bar{D}} \%. \quad (22)$$

A lower CV generally indicates a more repeatable response.

However, the coefficient of variation should be interpreted together with the absolute event density. Very weak responses can produce unstable percentage measures even when the absolute variation is small.

12 ON_PEAK Colour-Response Estimation

The current primary colour feature is based on ON-event density.

For each ROI and illumination channel, the maximum response across the repeated cycles is selected:

$$R_{\text{peak}} = \max_k D_{r,R,k}^{\text{ON}}, \quad (23)$$

$$G_{\text{peak}} = \max_k D_{r,G,k}^{\text{ON}}, \quad (24)$$

$$B_{\text{peak}} = \max_k D_{r,B,k}^{\text{ON}}. \quad (25)$$

The unnormalised response vector is

$$\mathbf{p}_r = [R_{\text{peak}}, G_{\text{peak}}, B_{\text{peak}}]. \quad (26)$$

The quantitative response vector is then normalised by the sum of the three components:

$$\mathbf{v}_r = \frac{\mathbf{p}_r}{R_{\text{peak}} + G_{\text{peak}} + B_{\text{peak}}}. \quad (27)$$

For any valid non-zero response,

$$v_R + v_G + v_B = 1. \quad (28)$$

This is the response vector exported as the current ON_PEAK colour measurement.

It is therefore **sum-normalised**.

Maximum-based normalisation is not used to calculate the quantitative response vector.

12.1 Rationale and Limitations of ON_PEAK

The ON_PEAK rule was retained because it provides a simple and interpretable separation between the responses to the three illumination channels.

It also allows the individual repeated measurements to remain available for diagnostic analysis rather than mixing ON, OFF and TOTAL measurements into one quantity.

However, selecting the maximum response has an important limitation: an unusually strong repetition can influence the final vector.

For this reason, the pipeline separately retains the response from every cycle and calculates repeatability statistics.

Future analysis could compare ON_PEAK with more robust alternatives such as:

- median response,
- trimmed mean,
- weighted averaging,
- robust statistical estimators,
- learning-based response models.

13 Display Colour Generation

The quantitative response vector is not itself an 8-bit RGB colour.

For visualisation, it is first rescaled relative to its largest component:

$$\mathbf{u}_r = \frac{\mathbf{v}_r}{\max(\mathbf{v}_r)}. \quad (29)$$

The base display brightness is controlled by

```
1 on_peak_display_brightness = 200.0
```

so that the initial display colour is

$$\mathbf{C}_r = B_{\text{display}} \mathbf{u}_r. \quad (30)$$

This maximum-based scaling is used only for visual rendering.

The exported quantitative vector $\mathbf{v}_r$ remains unchanged.

13.1 Display Desaturation

A small display-only desaturation is applied to reduce excessively vivid pseudo-colours.

Luminance is estimated using

$$Y = 0.2126R + 0.7152G + 0.0722B. \quad (31)$$

For desaturation fraction λ ,

$$\mathbf{C}_{r,\text{desat}} = (1 - \lambda)\mathbf{C}_r + \lambda[Y, Y, Y]. \quad (32)$$

The default configuration is

```
1 display_desaturation_fraction = 0.12
```

This transformation changes the visual output only.

14 Activity-Based Material Contrast

The pipeline also includes a separate display mechanism intended to increase visual contrast between a reference region and regions with substantially different event activity.

The default reference is

```
1 BRIGHTNESS_REFERENCE_ROI = "ROI_1"
```

For ROI r , total ON-event density across all RGB response windows is

$$A_r = \sum_{c,k} D_{r,c,k}^{\text{ON}}. \quad (33)$$

The activity ratio relative to the reference region is

$$q_r = \frac{A_r}{A_{\text{ref}}}. \quad (34)$$

A symmetric deviation is then defined as

$$d_r = |\ln(q_r)|. \quad (35)$$

The absolute logarithm treats both unusually high and unusually low event activity as deviations from the reference condition.

An activity similarity is calculated as

$$s_r = \exp(-\gamma d_r), \quad (36)$$

where γ is controlled by

```
1 BRIGHTNESS_CONTRAST_STRENGTH = 9.0
```

The display multiplier is

$$m_r = m_{\text{min}} + (1 - m_{\text{min}})s_r, \quad (37)$$

where

```
1 BRIGHTNESS_MIN_MULTIPLIER = 0.22
```

The final display colour becomes

$$\mathbf{C}_{r,\text{final}} = m_r \mathbf{C}_{r,\text{desat}}. \quad (38)$$

This brightness transformation is intentionally separated from the quantitative ON_PEAK vector.

It should be interpreted as a **display-level material-contrast enhancement**, not as calibrated surface brightness.

15 Pseudo-Colour Event Output

The final visualisation preserves the event-driven spatial structure of the sensor.

The software does not fill the complete ROI with a solid colour.

Instead, colour is applied only to pixels containing active events within the displayed temporal window.

The representation therefore combines

event spatial information + event temporal information + ROI response colour.

An optional persistence interval retains recently active event pixels for visualisation.

Persistence affects only the appearance of the output and does not modify the quantitative event measurements used for RGB-response estimation.

16 Usage and Configuration

16.1 Software Requirements

The current implementation requires:

- Python,
- NumPy,
- OpenCV,
- Matplotlib,
- Pillow.

Install the dependencies from the repository root using

```
python -m pip install -r requirements.txt
```

16.2 Selecting the Input Recording

Open

```
run_pipeline.py
```

and modify the input path:

```
1 DATA_FILE = Path(
2     r"path\to\recording.dat"
3 )
```

For example:

```
1 DATA_FILE = Path(
2     r"D:\event_data\experiment_01.dat"
3 )
```

Raw event-camera recordings are normally stored locally rather than committed to the public repository.

16.3 Running the Pipeline

From the repository root, execute

```
python run_pipeline.py
```

The main processing stages then run sequentially.

User interaction is required for reference-frame selection, ROI-count selection when `num_rois=None`, and polygonal ROI definition.

16.4 Changing the RGB Timing

The principal acquisition timing parameters are

```
1 FADE_DURATION_US = 1_000
2 PEAK_APPLY_TIME_US = 0
3 DARK_INTERVAL_US = 500
4 END_DELAY_US = 0
5 MEASURED_FLASH_PERIOD_US = None
```

These values should describe the acquisition experiment.

For example, if the physical illumination ramp lasted 500 μ s,

```
1 FADE_DURATION_US = 500
```

should be used.

If the adjacent-stage period has been independently measured, specify it directly:

```
1 MEASURED_FLASH_PERIOD_US = 1268
```

This is preferable to shifting individual RGB stages toward local event maxima.

16.5 Changing the Number of RGB Repetitions

The number of complete RGB cycles is configured through

```
1 TIMING = TimingConfig(
2     num_rgb_repeats=3,
3     ...
4 )
```

For five repetitions:

```
1 num_rgb_repeats=5
```

The program will then evaluate 15 RGB illumination stages.

16.6 Changing the Response Window

The current configuration includes

```
1 response_delay_us=0,
2 response_window_us=FADE_DURATION_US,
```

To measure only 500 μ s, starting 100 μ s after the predicted onset:

```
1 response_delay_us=100,
2 response_window_us=500,
```

The complete interval must remain within one illumination period.

16.7 Changing the Search Range

The flash-search configuration contains

```
1 FLASH_SEARCH = FlashSearchConfig(
2     first_red_search_duration_us=None,
3     ...
4 )
```

With `None`, the program may continue searching throughout the available recording.
For example,

```
1 first_red_search_duration_us=10_000_000
```

would restrict candidate first-onset search to approximately the first ten seconds after the configured read start.

16.8 Skipping an Initial Recording Segment

The default is

```
1 SKIP_BEFORE_TIME_US = 0
```

To begin processing two seconds after the first recorded event:

```
1 SKIP_BEFORE_TIME_US = 2_000_000
```

This option should normally remain zero unless the excluded section is known to be irrelevant.

16.9 Changing ROI Configuration

The default ROI configuration includes

```
1 ROI = ROIConfig(
2     num_rois=None,
3     nested_roi_mode="EXCLUDE_CONTAINED",
4     reference_window_us=50_000,
5     reference_step_us=10_000,
6     reference_search_duration_us=300_000,
7     reference_display_mode="TOTAL",
8     display_scale=1.5,
9     event_point_size=1,
10    automatic_reference_frame=False,
11    ...
12 )
```

Setting

```
1 num_rois=4
```

would always request four ROI polygons.

Changing

```
1 nested_roi_mode="INDEPENDENT"
```

disables automatic subtraction of completely contained regions.

Increasing

```
1 reference_window_us
```

can make sparse event structures easier to identify while drawing ROIs, but does not change the quantitative RGB response windows.

16.10 Changing Pseudo-Colour Appearance

The main colour-display parameters are

```
1 COLOUR = ColourConfig(
2     minimum_colour_response_events=1.0,
3     minimum_colour_response_density=1e-6,
4     on_peak_display_brightness=200.0,
5     display_desaturation_fraction=0.12,
6 )
```

Increasing

```
1 on_peak_display_brightness
```

makes the displayed events brighter.

Changing

```
1 display_desaturation_fraction
```

changes displayed saturation only.

Neither operation changes the exported ON_PEAK response vector.

16.11 Changing Material Contrast

The display-level activity contrast is configured using

```
1 BRIGHTNESS_REFERENCE_ROI = "ROI_1"
2 BRIGHTNESS_CONTRAST_STRENGTH = 9.0
3 BRIGHTNESS_MIN_MULTIPLIER = 0.22
```

A larger contrast strength causes ROIs whose activity differs from the reference to darken more rapidly.

The minimum multiplier controls the lowest permitted display brightness.

These values do not modify the quantitative response vector.

16.12 Changing GIF Settings

The default visualisation configuration includes

```
1 VIDEO = VideoConfig(
2     duration_us=500_000,
3     frame_window_us=500,
4     frame_step_us=500,
5     persistence_us=20_000,
6     display_mode="TOTAL",
7     gif_fps=15.0,
8     gif_max_frames=90,
9     ...
10 )
```

To display a longer sensor-time interval:

```
1 duration_us=1_000_000
```

To reduce the GIF size:

```
1 gif_max_frames=60
```

or increase

```
1 frame_step_us
```

The GIF frame rate changes playback speed only and does not change the event-camera timestamps.

17 Interactive ROI Operation

During reference-frame selection:

Key	Action
A / D	Move backward / forward in time
W / S	Increase / decrease accumulation duration
+ / -	Change display brightness scaling
M	Switch TOTAL / SIGNED / ON / OFF display
T	Toggle displayed event-point size
R	Return to an automatically selected high-coverage reference
Enter / Space	Confirm the current reference frame
Q / Esc	Cancel

During polygon definition:

Input	Action
Left mouse click	Add polygon vertex
Right mouse click / Z	Remove the most recent vertex
R	Clear and redraw the current polygon
Enter / Space	Confirm the current ROI
Q / Esc	Cancel

A polygon must contain at least three vertices.

18 Program Outputs

Each execution creates a new timestamped experiment directory under

```
new_led_full_recording_results/
```

The most important outputs are:

Output	Purpose
initial_activity.png	Coarse activity overview of the retained recording.
recording_summary.json	Whole-recording duration and event-activity information.
automatic_timing_report.json	Detailed diagnostic information from automatic first-phase detection.
automatic_timing_check.csv	Per-stage evidence used to evaluate the fixed RGB timing model.
flash_timing.csv	Predicted timing of the RGB stages and response windows.
flash_detection.png	Visual timing diagnostic for the selected first onset and subsequent fixed RGB windows.
roi_reference.png	Accumulated event reference used for ROI selection.
roi_effective_masks.png	Diagnostic representation of the statistical ROI masks.
flash_responses.csv	Raw and area-normalised ON, OFF and TOTAL responses for every ROI, colour and repetition.
diagnostic_response_vectors.csv	Mean ON/OFF/TOTAL response vectors retained for diagnostic analysis.

Output	Purpose
repeatability_summary.csv	Repeatability statistics across repeated RGB cycles.
normalised_colour_repeats.csv	Individual normalised ON-response vectors for each repetition.
response_vectors.csv	Final quantitative ON_PEAK response vectors.
analysis_metadata.json	Timing, ROI, brightness and experiment metadata for the completed run.
pseudo_colour_result.gif	Display-only pseudo-colour event animation.

19 Interpreting the RGB Response

Suppose an ROI produces

$$\mathbf{v} = [0.52, 0.32, 0.16]. \quad (39)$$

Within the current ON_PEAK definition, this means that approximately

- 52% of the three-channel peak response is associated with red illumination,
- 32% with green illumination,
- 16% with blue illumination.

It does not imply that the physical surface has the conventional RGB intensity

$$(0.52, 0.32, 0.16).$$

The measured response is influenced by the combined effects of

- illumination optical power,
- illumination spectrum,
- surface reflectance,
- event-sensor spectral sensitivity,
- contrast thresholds,
- temporal illumination behaviour,
- previous illumination state,
- motion,
- local scene structure,
- noise.

The response should therefore be interpreted as a relative colour-related event signature under the controlled experimental conditions.

20 Troubleshooting

20.1 DAT File Cannot Be Found

If the program reports that the input file does not exist, verify

```
1 DATA_FILE = Path("...")
```

and confirm that the path corresponds to an accessible local DAT file.

20.2 Automatic Timing Detection Fails

A timing-detection failure does not necessarily indicate a Python error.

Check:

1. whether the DAT contains the complete expected RGB sequence;
2. whether the configured fade duration matches the experiment;
3. whether the dark interval is correct;
4. whether the adjacent-stage period differs from the configured value;
5. whether sufficient pre-experiment background was recorded;
6. whether the assumed first physical illumination channel was actually red.

The timing diagnostics should be inspected before statistical thresholds are changed.

Reducing detection thresholds only until a candidate passes does not improve the physical synchronisation.

20.3 Later RGB Stages Gradually Drift

If the first stage appears plausible but later stages progressively move relative to the event response, the most likely cause is a mismatch between the configured period and the true hardware timing.

The preferred solution is to estimate or measure the true period and set

```
1 MEASURED_FLASH_PERIOD_US = ...
```

rather than independently moving each later RGB window.

20.4 Reference Image Is Too Sparse

Increase the reference accumulation window interactively using `W`, or increase

```
1 reference_window_us
```

in the ROI configuration.

This affects only the ROI reference display.

20.5 An ROI Contains Too Few Events

Possible causes include:

- the ROI is very small,
- the selected surface produces little activity,
- temporal alignment is incorrect,
- the response window is too short,
- the intended illumination did not reach the selected region.

An ROI should not be enlarged solely to obtain a visually preferable colour result. It should continue to represent the intended physical surface.

20.6 Contained Defect Is Included in the Reference

Confirm that

```
1 nested_roi_mode="EXCLUDE_CONTAINED"
```

is enabled and that the defect ROI is completely contained inside the outer polygon.

20.7 Pseudo-Colour Appears Too Dark

Display-only parameters that can be adjusted include

```
1 on_peak_display_brightness
2 BRIGHTNESS_CONTRAST_STRENGTH
3 BRIGHTNESS_MIN_MULTIPLIER
```

These values affect presentation and should not be interpreted as colour calibration parameters.

21 Limitations

21.1 Event Count Is Not Absolute Optical Intensity

An event camera responds to changes in logarithmic brightness.

Therefore,

$$\text{event count} \neq \text{absolute optical intensity.} \quad (40)$$

This is a fundamental limitation when attempting to infer colour from event activity alone.

21.2 Illumination Is Not Radiometrically Calibrated

The red, green and blue illumination channels may not produce equal optical power.

A larger response under one channel may therefore reflect both the target surface and the illumination source.

21.3 Sensor Spectral Sensitivity Is Not Calibrated

The monochrome event sensor may have wavelength-dependent sensitivity.

Without a calibrated sensor response model, measured differences between RGB channels cannot be attributed to surface reflectance alone.

21.4 PWM Generates Additional Temporal Structure

PWM-controlled illumination can introduce multiple brightness transitions within a physical illumination stage.

These transitions generate additional events and make direct peak-based timing interpretation difficult.

21.5 Motion and Illumination Both Generate Events

Events can be generated by illumination change, object motion or camera motion.

The current controlled pipeline does not completely separate these sources.

This becomes especially important when moving from static laboratory experiments toward dynamic scenes.

21.6 Colour Estimation Is ROI-Level

One response vector is currently estimated for each ROI.

That response is then applied to active event pixels within the same region.

The system therefore does not estimate an independent RGB response for every image pixel.

21.7 ON_PEAK Is Sensitive to Strong Repetitions

The maximum-response rule can be influenced by a single unusually strong cycle.

Repeatability outputs should therefore be inspected when assessing a result.

21.8 Timing Validation Is Not Hardware Synchronisation

Passing the automatic timing test means that the observed activity is statistically consistent with the configured fixed-period model.

It does not demonstrate that the event camera and illumination controller share a common clock or that every predicted onset is physically exact.

22 Future Work

22.1 Hardware Synchronisation

A direct digital timing signal between the illumination controller and the event-camera acquisition system would significantly reduce timing uncertainty.

The software would no longer need to infer the experiment start from event activity alone.

22.2 Illumination Calibration

The optical output of each red, green and blue channel could be measured experimentally.

This would allow the response to be corrected for unequal illumination power.

22.3 Sensor Spectral Calibration

The wavelength-dependent response of the event sensor could be characterised.

Combining illumination calibration with sensor spectral calibration would provide a more physically meaningful basis for colour estimation.

22.4 RGB Camera Ground Truth

A conventional RGB camera could record the same target surfaces and provide reference colour measurements.

The event-derived response could then be compared directly with frame-based ground truth.

This would allow quantitative evaluation rather than qualitative pseudo-colour inspection alone.

22.5 Quantitative Colour Evaluation

With calibrated ground truth, possible evaluation measures include:

- RGB reconstruction error,
- chromaticity error,
- perceptual colour difference,
- surface-colour classification accuracy,
- repeatability across different illumination conditions.

22.6 Robust Response Estimation

Alternative estimators could be compared with ON_PEAK, including:

- median response,
- trimmed mean,
- robust statistical estimators,
- uncertainty-weighted fusion,
- learning-based mappings.

22.7 Motion Compensation

Event-based optical flow, tracking or motion estimation could help separate motion-generated events from illumination-generated activity.

22.8 Dense Spatial Colour Estimation

Future work could replace one response vector per ROI with local or dense spatial estimation.

This would move the system toward more detailed pseudo-colour or event-based colour reconstruction.

22.9 Multimodal Sensor Fusion

A longer-term direction is the integration of

$$\text{Event Camera} + \text{RGB Camera} + \text{LiDAR}.$$

These sensors provide complementary information:

- event camera: high temporal resolution and dynamic visual information;
- RGB camera: dense appearance and conventional colour information;
- LiDAR: depth and three-dimensional geometry.

Such a combination could support robust low-latency perception for robotics and autonomous systems.

23 Conclusion

This project developed a complete processing framework for analysing the response of a monochrome event camera under sequential RGB illumination.

The pipeline combines memory-efficient DAT decoding, automatic first-phase localisation, fixed-period temporal alignment, interactive ROI selection, area-normalised event-response measurement, repeatability analysis, ON_PEAK response estimation and pseudo-colour visualisation.

A key methodological observation is that the largest event-count peak cannot automatically be treated as the physical illumination onset. PWM and other illumination dynamics may generate several strong event bursts during one illumination stage.

The implemented approach therefore detects one initial phase and uses a single timing model for the remaining RGB sequence instead of independently fitting every stage to the event data.

The quantitative RGB response is constructed from relative ON-event activity and should be interpreted as a sensor–illumination–surface response rather than calibrated surface colour. Display desaturation and activity-based brightness modification are applied only after the quantitative vector has been calculated.

The current framework provides a reproducible basis for further investigation of event-based colour response. Future improvements in hardware synchronisation, radiometric calibration, spectral modelling, RGB ground-truth evaluation, motion compensation and multimodal sensing could enable more rigorous and general colour estimation from event-camera data.

A Quick-Start Procedure

For a new recording:

1. Open `run_pipeline.py`.
2. Set the input DAT path:


```
1 DATA_FILE = Path(  
2     r"path\to\recording.dat"  
3 )
```

3. Confirm that the illumination timing parameters match the acquisition experiment.

4. Run:

```
python run_pipeline.py
```

5. Inspect the automatic timing result.

6. Select a suitable event reference frame.

7. Select the required number of ROIs and draw the ROI polygons.

8. Allow the quantitative analysis and pseudo-colour generation to complete.

9. Inspect `response_vectors.csv`, `repeatability_summary.csv`, and the timing diagnostics before interpreting the final pseudo-colour result.

B Notes on Reproducibility

Reproducing the software execution requires:

- the source code,
- the required Python packages,
- a compatible DAT recording.

Reproducing the exact physical experiment additionally requires knowledge of the acquisition conditions, including:

- event-camera configuration,
- RGB illumination order,
- illumination timing,
- LED drive behaviour,
- target surfaces,
- ROI definitions,
- sensor and object motion.

A successful execution of the software should therefore not by itself be interpreted as complete physical reproduction of the original experiment.